

Chapter 5A:

Algebra

8 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-ALG-C-01 generalises number properties to operate with algebraic expressions.

CHAPTER

5A

5.01 The commutative and associative laws

5.02 The distributive law

5.03 Variables and algebraic notation

5.04 Expressions from words

5.05 Substitution

5.07 Adding and subtracting terms

5.08 Multiplying terms

5.09 Dividing terms

How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL **Skill drill pages.** After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

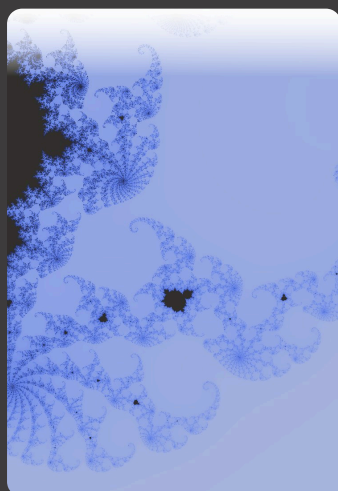
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
5.01	The commutative and associative laws	3		☐
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In this chapter you will

- use the commutative, associative and distributive laws
- use variables and algebraic notation
- write expressions from words
- substitute into expressions and formulas
- add, subtract, multiply and divide algebraic terms



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is Elephant Valley, the row of trunks along the right-hand edge of the main body. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
5.01

The commutative and associative laws

Date _____

Learning intentions

We are learning to:

- use the commutative and associative laws to make calculations easier

Success criteria

I can:

- ☐ know that order does not matter for + and $\times$ (commutative law)
- ☐ know that grouping does not matter for + and $\times$ (associative law)
- ☐ regroup numbers to make friendly pairs such as 10s and 100s

Key terms Write the definition as your teacher explains it.

Commutative

Associative

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate by pairing friendly numbers.

a $7 + 18 + 3$

b $91 + 18 + 9$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Commutative: $a + b = b + a$ and $a \times b = b \times a$.

ROUND 1 Questions 1–16: evaluate mentally. Pair the friendly numbers first.

1 $5 \times 26 \times 2$

2 $41 \times 25 \times 4$

3 $4 \times 34 \times 5$

4 $3 \times 50 \times 2$

5 $3 \times 5 \times 2$

6 $2 \times 50 \times 28$

7 $25 \times 4 \times 16$

8 $4 \times 4 \times 5$

9 $2 \times 26 \times 50$

10 $29 \times 2 \times 50$

11 $50 \times 2 \times 3$

12 $125 \times 8 \times 41$

13 $5 \times 22 \times 2$

14 $20 \times 40 \times 5$

15 $5 \times 2 \times 31$

16 $2 \times 34 \times 50$

YOU DO On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2 Questions 17–32: true or false?

17 $104 \div 8 = 8 \div 104$

18 $25 - 20 = 20 - 25$

19 $42 \div 6 = 6 \div 42$

20 $30 - 26 = 26 - 30$

21 $30 - 14 = 14 - 30$

22 $4 - 18 = 18 - 4$

23 $14 - 12 = 12 - 14$

24 $252 \div 21 = 21 \div 252$

25 $780 \div 30 = 30 \div 780$

26 $29 \times 28 = 28 \times 29$

27 $27 \times 13 = 13 \times 27$

28 $27 + 11 = 11 + 27$

29 $68 \div 4 = 4 \div 68$

30 $29 + 21 = 21 + 29$

31 $56 \div 4 = 4 \div 56$

32 $150 \div 25 = 25 \div 150$

Level 2 · Building up

Medium · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Evaluate by pairing friendly numbers.

a $2 \times 17 \times 5$

b $25 \times 31 \times 4$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

20 QUESTIONS

REMEMBER Associative: $(a + b) + c = a + (b + c)$.

ROUND 1 Questions 1–16: evaluate mentally. Rearrange to make tens first.

1 $72 + 32 + 28$

2 $64 + 15 + 6$

3 $41 + 44 + 29$

4 $11 + 30 + 19$

5 $31 + 64 + 39$

6 $61 + 68 + 29$

7 $35 + 45 + 15$

8 $31 + 34 + 39$

9 $76 + 19 + 24$

10 $40 + 51 + 10$

11 $33 + 84 + 7$

12 $25 + 41 + 35$

13 $24 + 19 + 6$

14 $22 + 87 + 8$

15 $16 + 53 + 34$

16 $51 + 36 + 39$

YOU DO On your own: **Medium** keep going: Round 2.

4 QUESTIONS MORE

ROUND 2 Questions 17–20: which law is shown? Write commutative, associative or neither.

17 $(2 \times 6) \times 5 = 2 \times (6 \times 5)$

18 $a + b = b + a$

19 $(a + b) + c = a + (b + c)$

20 $8 \times 7 = 7 \times 8$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate mentally.

a 13×30

b 7×400

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER Look for pairs making 10, 100 or 1000.

1 Does the commutative law work for subtraction? Test it with $10 - 4$.

2 Does the associative law work for division? Test it with $(24 \div 6) \div 2$.

3 Evaluate $125 \times 17 \times 8$ mentally.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Evaluate $16 + 27 + 4$.

Level 2

Evaluate $5 \times 23 \times 2$.

Level 3

Evaluate $20 \times 13 \times 5$.

 Extension · Stretch yourself Worked example

Evaluate $4 \times 7 \times 25 \times 3 \times 5 \times 2$ mentally.

1. Regroup: $(4 \times 25) \times (5 \times 2) \times 7 \times 3$
2. $100 \times 10 \times 21$

Answer: 21 000

YOU DO

On your own: read the worked example, then try these.

- E1** Evaluate $8 \times 125 \times 9 \times 5 \times 2$ mentally.

- E2** Explain why $38 + 47 + 62 + 53$ is easy to do mentally.

Lesson summary

Missed the lesson? Start here.

Remember

1. Commutative: $a + b = b + a$ and $a \times b = b \times a$.
2. Associative: $(a + b) + c = a + (b + c)$.
3. Look for pairs making 10, 100 or 1000.
4. Not true for $-$ and $\div$.

 Level 1

$$7 + 18 + 3$$

1. $(7 + 3) + 18$

Answer: 28

 Level 2

$$25 \times 31 \times 4$$

1. $(25 \times 4) \times 31$

Answer: 3100

 Level 3

$$13 \times 30$$

1. $13 \times 3 \times 10$

Answer: 390

 Extension

$$4 \times 7 \times 25 \times 3 \times 5 \times 2$$

1. $100 \times 10 \times 21$

Answer: 21 000

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LESSON
5.02

The distributive law

Date _____

Learning intentions

We are learning to:

- use the distributive law for mental arithmetic

Success criteria

I can:

- ☐ split a number into tens and units, or a friendly number $\pm$ a bit
- ☐ multiply each part, then add or subtract
- ☐ choose the easiest way to split

Key terms Write the definition as your teacher explains it.

Distributive law

Expand

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the distributive law.

a 22×9 as $22 \times (10 - 1)$

b 17×8 as $(10 + 7) \times 8$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER $a \times (b + c) = a \times b + a \times c$.

ROUND 1 Questions 1–16: evaluate by splitting the number: $6 \times 23 = 6 \times 20 + 6 \times 3$.

1 3×21

2 5×95

3 3×37

4 5×64

5 4×68

6 9×16

7 6×97

8 3×49

9 6×63

10 5×99

11 9×92

12 9×84

13 9×15

14 9×37

15 5×28

16 8×81

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 9–16: expand the brackets.

9 $4(n - 9)$

10 $4(t - 4)$

11 $5(c - 7)$

12 $4(y - 3)$

13 $4(k - 2)$

14 $8(k - 11)$

15 $2(c - 6)$

16 $2(t + 6)$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use the distributive law.

a 48×6

b 99×35

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER Or near a friendly number: $9 = 10 - 1$.

1 Evaluate $7 \times 68 + 7 \times 32$ mentally.

2 Evaluate $15 \times 43 - 15 \times 33$ mentally.

3 Show two different ways to evaluate 18×25 .

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Evaluate 12×7 using the distributive law.

Level 2

Evaluate 19×6 .

Level 3

Evaluate 102×45 .

 Extension · Stretch yourself Worked example

Evaluate 999×47 mentally.

1. $999 = 1000 - 1$
2. $47\ 000 - 47$

Answer: 46 953

YOU DO

On your own: read the worked example, then try these.

- E1** Evaluate 1001×64 mentally.

- E2** Evaluate $37 \times 99 + 37$ mentally.

Lesson summary

Missed the lesson? Start here.

Remember

1. $a \times (b + c) = a \times b + a \times c$.
2. Split into tens and units: $17 = 10 + 7$.
3. Or near a friendly number: $9 = 10 - 1$.
4. Multiply each part, then add or subtract.

 Level 1

$$22 \times 9$$

1. $22 \times 10 - 22$

Answer: 198

 Level 2

$$17 \times 8$$

1. $80 + 56$

Answer: 136

 Level 3

$$99 \times 35$$

1. $3500 - 35$

Answer: 3465

 Extension

$$999 \times 47$$

1. $47\ 000 - 47$

Answer: 46 953

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LESSON
5.03

Variables and algebraic notation

Date _____

Learning intentions

We are learning to:

- use a variable to describe a pattern
- write and read algebraic notation

Success criteria

I can:

- ☐ write a general rule using a variable
- ☐ simplify notation, e.g. $5 \times a \times 2 = 10a$
- ☐ write a term in expanded form

Key terms Write the definition as your teacher explains it.

Variable

Pronumeral

Term

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write the rule, then simplify.

- a** Write the rule for: $3 - (-3) = 2 \times 3$, $5 - (-5) = 2 \times 5$,
 $11 - (-11) = 2 \times 11$

Your teacher works this out. Copy every step.

- b** Simplify $x + x + x + x$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER A **VARIABLE** is a letter that stands for a number.

ROUND 1 Questions 1–16: simplify. Write the number first, then the letters in alphabetical order.

1 $b \times 8$

2 $a \times 11$

3 $y \times p \times 7$

4 $p \times 4$

5 $t \times 6$

6 $a \times a$

7 $m \times m$

8 $y \times k \times 12$

9 $k \times 4$

10 $11 \times a$

11 $6 \times c$

12 $9 \times a$

13 $p \times p$

14 $6 \times a$

15 $k \times k$

16 $m \times 3$

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 17–22: write in expanded form.

17 $4b^2p$

18 $6p^2t$

19 $4a^2x$

20 $3b^2m$

21 $5k^3p$

22 $3k^2m$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write in expanded form.

a Write $4mnk$ in expanded form.

b Write $9c^2$ and $\frac{6a}{b}$ in expanded form.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Patterns and rules.

3 QUESTIONS

REMEMBER $a \times a = a^2$; $a \div 5 = a/5$.

1 Write a rule for: $4 + 4 = 2 \times 4$, $7 + 7 = 2 \times 7$, $10 + 10 = 2 \times 10$

2 Write a rule for: $3 \times 0 = 0$, $8 \times 0 = 0$, $15 \times 0 = 0$

3 Matchstick squares: 1 square uses 4 sticks, 2 squares use 7, 3 squares use 10. Find a rule for n squares.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Simplify $y + y + y$.

Level 2

Simplify $4 \times p \times 3 \times q$.

Level 3

Write $5a^2b$ in expanded form.

Extension · Stretch yourself**Worked example**

Write a rule for the pattern: $1 \times 3 = 2^2 - 1$, $2 \times 4 = 3^2 - 1$, $3 \times 5 = 4^2 - 1$

1. First number is n , second is $n + 2$
2. The square is of $n + 1$

Answer: $n(n + 2) = (n + 1)^2 - 1$

YOU DO

On your own: read the worked example, then try these.

- E1** Write a rule for: $5 \times 5 - 4 \times 6 = 1$, $8 \times 8 - 7 \times 9 = 1$, $3 \times 3 - 2 \times 4 = 1$

- E2** Test your rule with a number of your own.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A **variable** is a letter that stands for a number.
 2. Leave out $\times$: $3 \times a = 3a$. Number first, letters in order.
 3. $a \times a = a^2$; $a \div 5 = a/5$.
 4. Expanded form shows every $\times$.

Level 1

Rule for $3 - (-3) = 2 \times 3 \dots$

1. Replace the number with a

Answer: $a - (-a) = 2a$

Level 2

$5 \times a \times 2$

1. $5 \times 2 = 10$

Answer: $10a$

Level 3

$2 \times m \times w \times 10 \times w$

1. $20 \times m \times w^2$

Answer: $20mw^2$

Extension

$9c^2$

1. $9 \times c \times c$

Answer: $9 \times c \times c$

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LESSON
5.04

Expressions from words

Date _____

Learning intentions

We are learning to:

- write algebraic expressions from worded statements

Success criteria

I can:

- ☐ know the words for +, −, × and ÷ (sum, difference, product, quotient)
- ☐ write an expression using a variable
- ☐ use brackets when needed

Key terms Write the definition as your teacher explains it.

Expression

Consecutive

Notes

Level 1 · Getting started

Easy · 24 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write an expression using N .

a 3 times the number N

b the difference between N and 5

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS**

REMEMBER Sum → +, difference → −.

ROUND 1 Questions 1–12: write an expression.

1 b decreased by 2

2 double c

3 c minus 3

4 c decreased by 4

5 7 less than c

6 c minus 2

7 5 less than c

8 a minus 3

9 a increased by 10

10 half of a

11 b divided by 7

12 a minus 4

YOU DO On your own: **Medium** keep going: Round 2.

4 QUESTIONS MORE

ROUND 2 Questions 13–16: write an expression for each amount.

13 the cost of y pens at \$13 each

14 \$70 shared equally by k people

15 the number of legs on k spiders

16 y minutes plus 10 minutes

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write an expression.

a $2H$ decreased by k

b double the sum of x and 4

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Write an expression for each situation.

3 QUESTIONS

REMEMBER More than $\rightarrow +$, less than $\rightarrow -$ (order!).

1 The cost of n pens at \$3 each.

2 Sam is s years old. His sister is 4 years older. Find the sum of their ages.

3 A rectangle is L cm long and 5 cm wide. Write its perimeter.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write “the number plus 12” using N .

Level 2

Write “4 less than y ”.

Level 3

Write “the product of m and 5, then add 2”.



Extension · Stretch yourself



Worked example

Write an expression for the sum of three consecutive numbers starting at n , and simplify it.

1. $n + (n + 1) + (n + 2)$
2. Collect like terms

Answer: $3n + 3$

YOU DO

On your own: read the worked example, then try these.

- E1 Write and simplify the sum of three consecutive even numbers starting at e .

- E2 Explain why the sum of 3 consecutive numbers is always divisible by 3.

Lesson summary

Missed the lesson? Start here.

Remember

1. Sum $\rightarrow +$, difference $\rightarrow -$.
2. Product $\rightarrow \times$, quotient $\rightarrow \div$.
3. More than $\rightarrow +$, less than $\rightarrow -$ (order!).
4. Use brackets when a whole group is multiplied.

Level 1

3 times N

1. $3 \times N$

Answer: $3N$

Level 2

N decreased by 5

1. $N - 5$

Answer: $N - 5$

Level 3

Double the sum of x and 4

1. Brackets first

Answer: $2(x + 4)$

Extension

3 consecutive numbers

1. $n + n + 1 + n + 2$

Answer: $3n + 3$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.05

Substitution

Date _____

Learning intentions

We are learning to:

- find the value of an expression by substituting numbers

Success criteria

I can:

- ☐ replace each variable with its value, using brackets
- ☐ follow the order of operations
- ☐ substitute into a formula

Key terms Write the definition as your teacher explains it.

Substitute

Evaluate

Formula

Notes

Level 1 · Getting started

Easy · 24 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Substitute $a = 2$, $b = 5$ and $c = 6$.**a** If $a = 2$ and $b = 5$, find $a + b$.

Your teacher works this out. Copy every step.

b If $a = 2$ and $c = 6$, find $3a - c$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS****REMEMBER** Write the expression.**ROUND 1** Questions 1–12: substitute, then evaluate.**1** $p + 3$, when $p = 2$ **2** $9x + 5$, when $x = 5$ **3** $5n$, when $n = 8$ **4** $8p$, when $p = 3$ **5** $8x$, when $x = 10$ **6** $5a$, when $a = 9$ **7** $p + 9$, when $p = 5$ **8** $p - 7$, when $p = 5$ **9** $3p$, when $p = 9$ **10** $4p + 1$, when $p = 12$ **11** $7p + 3$, when $p = 6$ **12** $n + 4$, when $n = 12$

YOU DO On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2 Questions 13–24: if $a = 4$ and $b = 7$, evaluate.

13 $10 - a$

14 $4b - 3a$

15 $3b$

16 $b - a$

17 $2ab$

18 $ab - 10$

19 b^2

20 $a + b$

21 $a + b + 9$

22 $2a + b$

23 ab

24 $5a - b$

Level 2 · Building up

Medium · 22 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Substitute, then use the formula $C = 30h + 19$.

a If $a = 2$, $b = 5$, $c = 6$, find abc .

b If $C = 30h + 19$, find C when $h = 3$.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER Replace each variable with its value in brackets.

ROUND 1 Questions 1–16: substitute, then evaluate. Take care with negatives.

1 $x^2 + x$, $x = 3$

2 $10 - x$, $x = -1$

3 $4 - 2x$, $x = -4$

4 $10 - x$, $x = 2$

5 $10 - x$, $x = 1$

6 $3x$, $x = 2$

7 $x^2 + x$, $x = -6$

8 $4 - 2x$, $x = 1$

9 $5x + 1$, $x = -5$

10 x^2 , $x = 2$

11 $3x$, $x = 3$

12 $x + 8$, $x = -6$

13 $5x + 1$, $x = -2$

14 $x^2 + x$, $x = -2$

15 x^2 , $x = 5$

16 $2x - 5$, $x = 5$

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 17–22: substitute into the formula.

17 $s = d \div t$, $d = 108$, $t = 9$

18 $A = lw$, $l = 11$, $w = 11$

19 $C = 30h + 19$, $h = 6$

20 $P = 2l + 2w$, $l = 14$, $w = 2$

21 $P = 2l + 2w$, $l = 15$, $w = 3$

22 $s = d \div t$, $d = 90$, $t = 9$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Substitute into $C = 30h + 19$.

a Find C when $h = -1$.

Your teacher works this out. Copy every step.

b Find C when $h = 2.5$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Substitute into each formula.

3 QUESTIONS

REMEMBER Use the order of operations.

1 $P = 2L + 2W$. Find P when $L = 8$ and $W = 3$.

2 $A = \frac{1}{2}bh$. Find A when $b = 10$ and $h = 7$.

3 $F = 1.8C + 32$. Find F when $C = 25$.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

If $m = 4$, evaluate $5m$.

Level 2

If $x = 3$ and $y = 7$, evaluate $2x + y$.

Level 3

If $A = l^2 - 3l$, find A when $l = -2$.

Extension · Stretch yourself**Worked example**

A taxi charges $C = 4 + 2.5k$ dollars for k km. How far can you go for \$39?

1. $39 = 4 + 2.5k$
2. $35 = 2.5k$
3. $k = 35 \div 2.5$

Answer: 14 km

YOU DO

On your own: read the worked example, then try these.

- E1** Using the same formula, find the cost of a 22 km trip.

- E2** A phone plan costs $C = 15 + 0.2m$ for m minutes. How many minutes for \$27?

Lesson summary

Missed the lesson? Start here.

Remember

1. Write the expression.
2. Replace each variable with its value in brackets.
3. Use the order of operations.
4. Remember $3a$ means $3 \times a$.

Level 1

$$a + b, a = 2, b = 5$$

1. $2 + 5$

Answer: 7

Level 2

$$3a - c$$

1. $3 \times 2 - 6$

Answer: 0

Level 3

$$C = 30h + 19, h = -1$$

1. $30 \times (-1) + 19$

Answer: -11

Extension

Taxi for \$39

1. $2.5k = 35$

Answer: 14 km

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LESSON
5.07

Adding and subtracting terms

Date _____

Learning intentions

We are learning to:

- simplify expressions by collecting like terms

Success criteria

I can:

- ☐ identify **like terms** (same letters, same powers)
- ☐ add and subtract the coefficients of like terms
- ☐ collect several sets of like terms

Key terms Write the definition as your teacher explains it.

Like terms

Coefficient

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Simplify by collecting like terms.

a $3m + 4m$

b $12m - 4m$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER **LIKE TERMS** have exactly the same letters and powers.

ROUND 1 Questions 1–16: simplify by adding or subtracting like terms.

1 $10n - 5n$

2 $6p - 8p$

3 $3t + t$

4 $c + 12c$

5 $4y - 2y$

6 $k + 11k$

7 $3m + 6m$

8 $12p - 6p$

9 $12t + 9t$

10 $12x - 7x$

11 $2m - 3m$

12 $6y - 2y$

13 $5c - 7c$

14 $7a - a$

15 $5p - 10p$

16 $7x - 11x$

YOU DO On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2 Questions 17–32: like terms? Write yes or no.

17 nt and $9tn$

18 tx and $4xt$

19 bc and $9cb$

20 $2n$ and $9n$

21 bx and $9xb$

22 $7a$ and $3n$

23 $7p$ and $3x$

24 c^2 and $8c$

25 $5m$ and $8m$

26 kn and $6nk$

27 $8t$ and $8t$

28 $5m$ and $9m$

29 p^2 and $8p$

30 ap and $3pa$

31 cy and $8yc$

32 $9a$ and $4x$

Level 2 · Building up

Medium · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Simplify by collecting like terms.

a $7y - 4y - y$

Your teacher works this out. Copy every step.

b $6uv + 4 + 12uv$

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

18 QUESTIONS

REMEMBER Add or subtract only like terms.

ROUND 1 Questions 1–12: simplify. Collect the like terms.

1 $c - 2y - 7c$

2 $b - 7k - 3b$

3 $8a - 7b - a + 9$

4 $7c - 9p - c$

5 $7m + 4n + 3m + 5$

6 $5n - 9x + 7n$

7 $n - 8t + 2n$

8 $3a - 8k + 9a$

9 $9c - 8y - 5c$

10 $7k - p - 4k + 9$

11 $b + x + 5b + 6$

12 $c - 4y + 5c$

YOU DO

On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2

Questions 13–18: simplify.

13 $9mp + 6m^2 - 4mp + 9m^2$

14 $8mp - 2m^2 - 9mp + 6m^2$

15 $7mt - 8m^2 - 7mt + m^2$

16 $8cp - 4c^2 - 4cp + 5c^2$

17 $2bt - 4b^2 - 7bt + 7b^2$

18 $7bt - b^2 - 2bt + 7b^2$



Level 3 • Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Simplify. Collect each kind of term separately.

a $5h^2 + 6h + 8h^2 + 11h$

b $13h^3 - 6h - 8h - 11h^3$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Simplify.

3 QUESTIONS

REMEMBER Keep the sign in front of each term.

$$\textcircled{1} \quad 5h^2 + 6h + 8h^2 + 11h$$

$$\textcircled{2} \quad 13h^3 - 6h - 8h - 11h^3$$

③ $3ab - 2a + 5ab + 7a - b$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name	Class
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Simplify $6k + 3k$.



Simplify $4a + 7 - a + 2$.



Simplify $3x^2 + 5x - x^2 - 8x$.

Extension · Stretch yourself**Worked example**Simplify $4x^2y - 3xy^2 + x^2y + 5xy^2 - xy$.

1. Like terms: $4x^2y + x^2y$
2. $-3xy^2 + 5xy^2$
3. $-xy$ has no partner

Answer: $5x^2y + 2xy^2 - xy$ **YOU DO**

On your own: read the worked example, then try these.

- E1**
- Simplify
- $7a^2 - 3a + 2 - a^2 + 5a - 8$
- .

- E2**
- Find the missing term:
- $5m + ? - 2m = 9m$

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Like terms have exactly the same letters and powers.
 2. Add or subtract only like terms.
 3. Keep the sign in front of each term.
 4. Write the answer with the highest power first.

Level 1

$$3m + 4m$$

1. $3 + 4 = 7$

Answer: $7m$ **Level 2**

$$7y - 4y - y$$

1. $7 - 4 - 1$

Answer: $2y$ **Level 3**

$$6uv + 4 + 12uv$$

1. $6 + 12 = 18$

Answer: $18uv + 4$ **Extension**

$$5h^2 + 6h + 8h^2 + 11h$$

1. h^2 terms, then h terms

Answer: $13h^2 + 17h$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.08

Multiplying terms

Date _____

Learning intentions

We are learning to:

- multiply algebraic terms

Success criteria

I can:

- ☐ multiply the numbers first, then the letters
- ☐ write repeated letters as powers
- ☐ use the sign rules for negative terms

Key terms Write the definition as your teacher explains it.

Coefficient

Power

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Simplify.

a $3m \times 4m$

b $a \times a \times a \times a$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Multiply the numbers (coefficients).

ROUND 1 Questions 1–16: simplify. Multiply the numbers, then the letters.

1 $7k \times 7p$

2 $5 \times 9t$

3 $2b \times 8c$

4 $3b \times 8y$

5 $4 \times 5c$

6 $7b \times 8n$

7 $3c \times 2p$

8 $8c \times 4p$

9 $5 \times 3a$

10 $4 \times 4m$

11 $9 \times 4k$

12 $5a \times 3c$

13 $6 \times 4k$

14 $3 \times 6p$

15 $8a \times 7b$

16 $4a \times 8c$

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 17–24: simplify. Take care with the signs.

17 $-6c \times (-3n)$

18 $-4t \times 6y$

19 $-3a \times (-5n)$

20 $6a \times (-5k)$

21 $-6k \times 3m$

22 $5b \times 3n$

23 $-6m \times (-2n)$

24 $x \times (-4y)$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Simplify. Watch the signs.

a $-7d \times (-3d)$

Your teacher works this out. Copy every step.

b $lmn \times lmn \times lmn$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Simplify.

3 QUESTIONS

REMEMBER Same letter repeated → use a power.

1 $3pq \times 5pq$

2 $lmn \times lmn \times lmn$

3 $2a^2 \times 3a^3$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Simplify $5 \times 3k$.

Level 2

Simplify $4a \times 6ab$.

Level 3

Simplify $-2x \times 3x \times (-y)$.

Extension · Stretch yourself**Worked example**Simplify $(2ab)^3$.

- $2ab \times 2ab \times 2ab$
- Numbers: $2 \times 2 \times 2 = 8$
- a^3b^3

Answer: $8a^3b^3$ **YOU DO**

On your own: read the worked example, then try these.

E1 Simplify $(-3m)^2$.**E2** Find the missing term: $4x \times ? = 20x^2y$.**Lesson summary**

Missed the lesson? Start here.

Remember
1. Multiply the numbers (coefficients).
3. Same letter repeated $\rightarrow$ use a power.2. Multiply the letters, in alphabetical order.
4. Signs: same $\rightarrow$ +, different $\rightarrow$ -.**Level 1**

$3m \times 4m$

1. $3 \times 4 = 12$, $m \times m = m^2$

Answer: $12m^2$ **Level 2**

$9v \times 4 \times 5v$

1. $9 \times 4 \times 5 = 180$

Answer: $180v^2$ **Level 3**

$-7d \times (-3d)$

1. $- \times - = +$

Answer: $21d^2$ **Extension**

$(2ab)^3$

1. $8, a^3, b^3$

Answer: $8a^3b^3$

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LESSON
5.09

Dividing terms

Date _____

Learning intentions

We are learning to:

- divide algebraic terms

Success criteria

I can:

- ☐ write a division as a fraction
- ☐ cancel common factors in the numbers and the letters
- ☐ simplify answers that are still fractions

Key terms Write the definition as your teacher explains it.

Quotient

Cancel

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Simplify.

a $15m \div 3$

b $9v \div v$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Write the division as a fraction.

ROUND 1 Questions 1–16: simplify.

1 $72p \div 8$

2 $8a \div 4$

3 $24t \div 6$

4 $54n \div 9$

5 $18t \div 6$

6 $35m \div 7$

7 $32k \div 8$

8 $18b \div 3$

9 $18x \div 9$

10 $49n \div 7$

11 $49x \div 7$

12 $25m \div 5$

13 $6p \div 2$

14 $48c \div 8$

15 $56c \div 7$

16 $9y \div 3$

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 17–24: simplify the fraction.

17 $\frac{24p^2}{6p}$

18 $\frac{6n}{2n}$

19 $\frac{27n}{3n}$

20 $\frac{42t^2}{6t}$

21 $\frac{35mt}{5m}$

22 $\frac{40x}{5x}$

23 $\frac{4y^2}{2y}$

24 $\frac{18nx}{3x}$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Simplify.

a $32de^2 \div 10de$

b $\frac{-3 \times 2 \times y \times 2x}{3x \times (-x)}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Simplify.

3 QUESTIONS

REMEMBER Cancel letters that are on top and bottom.

1 $\frac{-3 \times 2 \times y \times 2x}{3x \times (-x)}$

2 $\frac{4a \times 6b}{8ab}$

3 $\frac{10m^2n}{2m \times 5n}$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Simplify $21p \div 7$.

Level 2

Simplify $\frac{16ab^2}{4b}$.

Level 3

Simplify $\frac{-10 \times 3}{15x}$.

Extension · Stretch yourself**Worked example**Simplify $\frac{6a3b \times 4ab2}{12a2b2}$.

1. Top: $24a^4b^3$

2. $\div 12a^2b^2$

Answer: $2a^2b$ **YOU DO**

On your own: read the worked example, then try these.

E1 Simplify $\frac{(3m)2}{6m}$.

E2 Simplify $\frac{8 \times 2y \times 3y}{6xy \times 2x}$.

Lesson summary

Missed the lesson? Start here.

Remember 1. Write the division as a fraction.
3. Cancel letters that are on top and bottom.2. Cancel the numbers by their HCF.
4. Signs: same $\rightarrow +$, different $\rightarrow -$.**Level 1**

$15m \div 3$

1. $15 \div 3 = 5$

Answer: $5m$ **Level 2**

$\frac{12 \times 2y}{4xy}$

1. $12 \div 4 = 3$, cancel x and y

Answer: $3x$ **Level 3**

$32de^2 \div 10de$

1. $32/10 = 16/5$, cancel d and e

Answer: $\frac{16e}{5}$ **Extension**

$\frac{6a3b \times 4ab2}{12a2b2}$

1. $24 \div 12 = 2$

Answer: $2a^2b$

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